Team: Netherite

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Evaluating Multilingual Reasoning through a Chain-of-Thought Lens

Previous Work + Gap

Previous work has done:

- Shown that Chain-of-Thought (CoT) prompting improves reasoning, particularly on mathematical and symbolic tasks (Wei et al., 2022; Sprague et al., 2025).

- Demonstrated that irrelevant information can disrupt CoT reasoning, **in English** (Shi et al., 2023a; Mirzadeh et al., 2025).

- Introduced multilingual temporal reasoning benchmarks such as TRD, **but without evaluating CoT prompting** (Mazzia et al., 2026) (Temporal reasoning is the ability to understand and reason about time, dates, durations, and the order of events.)

Research Gap: No prior work jointly examines how reasoning language, CoT prompting, and distractors interact in multilingual temporal reasoning, or systematically compares English and Hindi reasoning trace structures.

Research Questions

RQ1: Does CoT improve temporal reasoning accuracy, and how does this vary across languages and difficulty levels?

RQ2: Does CoT increase susceptibility to distractors, and does this depend on distractor type and language?

RQ3: How does reasoning in Hinglish affect accuracy and reasoning trace structure compared to English and Hindi? (To Do)

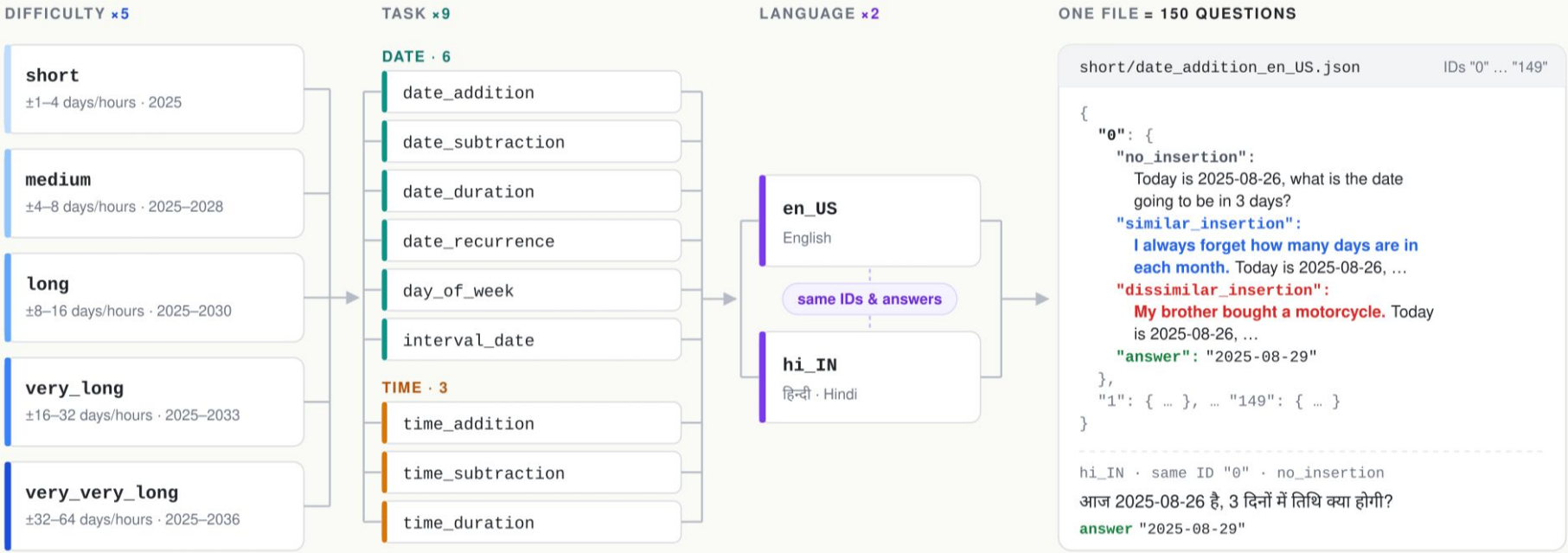
RQ4: Can cross-lingual prompting (CLP) improve Hindi reasoning accuracy and bring its reasoning process closer to English? (To Do)

Dataset structure

seed 9 · 150 questions per file

`dataset/<difficulty>/<task>_<lang>.json`

`config/generator.cfg` · `custom-generator.py`



Pipeline so far

as of 2026-10-16

`dataset → responses → scores → statistics`

01 GENERATE

`TRD library (unmodified)`

`config/generator.cfg`

custom-generator.py

- extends TRD's BaseExperiment class
- same random state per language → parallel en/hi questions
- adds similar & dissimilar distractor sentences

Dataset/

90 files · 13,500 questions
× 3 insertion variants

`repo root`

02 QUERY

`config/prompt.yaml`

`config/sarvam.yaml`

sarvam.py

- 2 modes: direct · cot
- answer in the target language, end with "Answer: …"
- sarvam-105b · temp 0
  6 concurrent requests
- retries; resumes on rerun

results.json ×4

40,244 of 81,000 responses
paused: API credits ran out

`experiments/2026-09-29_run-*`
`experiments/2026-09-30_run-*`

03 MERGE

`task filter per difficulty`

build_merged.py

- keeps a question only if all 12 responses exist:
- 2 languages × 3 insertions × 2 modes
- → every comparison is paired on the same question

merged_results.json

37,068 responses
3,089 complete questions

`experiments/2026-10-01_analysis-rahul`

04 SCORE

`TRD weekday names`

metrics.py

- parse last "Answer:" line (उत्तर, Devanagari digits ok)
- exact match vs gold answer
- CoT target-language ratio (Hindi vs English words)

by_*.csv

accuracy + CoT language ratio by difficulty · lang · mode · insertion · task

`experiments/2026-10-01_metrics-breakdown-rahul`

05 ANALYSE

`merged data → build_responses.py`

`metrics.py · temporal_answers.py`

analysis.py

- cluster-bootstrap 95% CIs (2,000 question resamples)
- McNemar + Holm correction
- Mantel–Haenszel odds ratios
- writes tables, figures, numbers

outputs/

5 LaTeX tables · 3 PDF figures
249 numbers used in the text of the mid-report (report/main.tex)

`experiments/2026-10-02_mid-report-analysis`

Exploratory pass · Oct 1 not used by the report — `experiments/2026-10-01_analysis-rahul`

7 scripts: a first look at CoT deltas, error types, distractor reuse, trace faithfulness and en/hi agreement.
No significance tests. Superseded by analysis.py, which reuses its temporal_answers.py and reproduces its counts.

config / dependency · script · data it writes · exploratory, superseded

Still to run: **very_very_long** and the remaining ~50% of short · medium · long

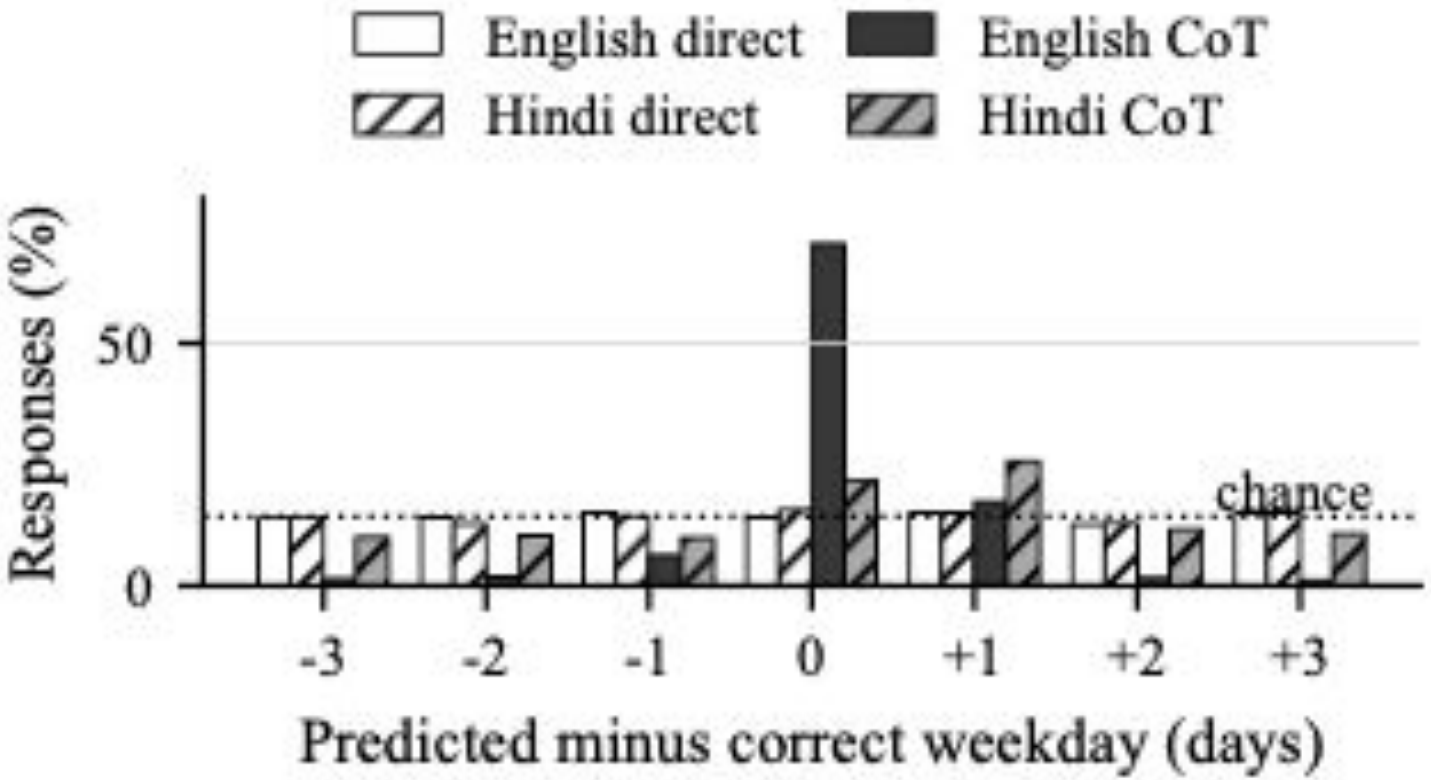
Each run keeps copies of the scripts and configs it ran in its own `experiments/<date>_<name>/` folder.

Results

- 3,089 questions answered in all 12 conditions (46% of the 6,750 planned)
- Runs stopped when the API credit ran out, so the very very long level is not run yet

CoT Improves Temporal Reasoning, but Unequally Across Languages (RQ1)

- English: Accuracy increases from 63.2% (Direct) to 91.1% (CoT), a gain of +27.9 percentage points.
- Hindi: Accuracy increases from 57.7% to 62.9%, a much smaller gain of +5.1 points.
- CoT improves performance across all 9 tasks in English, but only 5 tasks in Hindi.
- Hindi CoT worsens performance on date subtraction (92.2% → 78.2%) and interval date (35.1% → 17.9%).
- Key Finding: CoT provides substantial benefits in English, but its effectiveness does not transfer equally to Hindi.

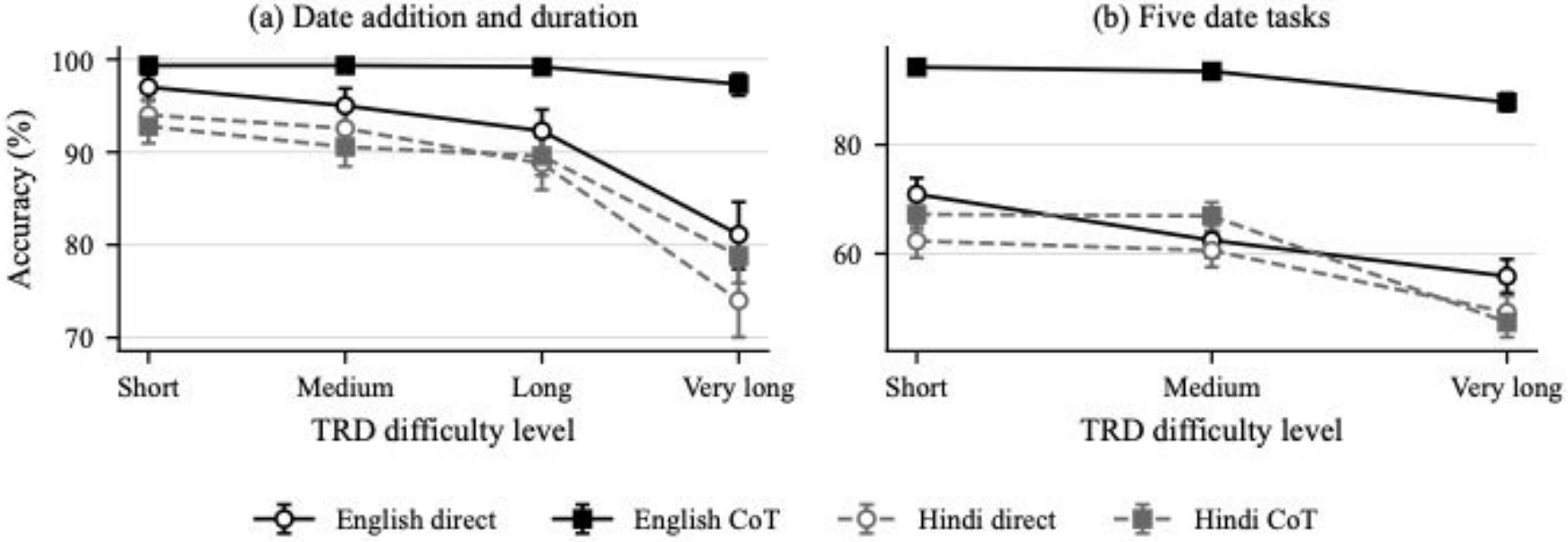

English direct
English CoT
Hindi direct
Hindi CoT
Responses (%)
50
0
chance
-3
-2
-1
0
+1
+2
+3
Predicted minus correct weekday (days)

CoT Performance Varies by Task and Difficulty (RQ1)

- Day of Week: English improves from 14.1% → 70.5% with CoT, while Hindi improves only from 15.7% → 21.8%.
- Date Recurrence: English improves from 24.5% → 92.4%, compared to 4.8% → 26.0% in Hindi.
- The English–Hindi accuracy gap increases with difficulty, particularly under CoT.
- Across five date tasks, the CoT gap grows from 27.0 to 40.4 points, while the direct-answer gap stays below 9 points.
- Key Finding: CoT amplifies the English–Hindi performance gap, especially on difficult temporal reasoning tasks.

Distractors Disproportionately Affect Hindi CoT (RQ2)

- English: Distractors reduce accuracy by at most 1.2 percentage points, regardless of prompting method.
- Hindi Direct: Accuracy remains largely unaffected by distractors.
- Hindi CoT accuracy:
  - No distractor: 65.2%
  - Time-related distractor: 60.7% (−4.5 points)
  - Unrelated distractor: 62.7% (−2.5 points)
- The additional distraction penalty under CoT is 5.2 points for time-related and 2.8 points for unrelated distractors in Hindi.
- Key Finding: CoT increases susceptibility to distractors in Hindi, particularly those containing temporal information.


(a) Date addition and duration
Accuracy (%)
100
90
80
70
Short
Medium
Long
Very long
TRD difficulty level
(b) Five date tasks
80
60
Short
Medium
Very long
TRD difficulty level
English direct
English CoT
Hindi direct
Hindi CoT

Distractor Incorporation into Reasoning Traces (RQ2)

Hindi CoT traces mention distractors substantially more frequently than English traces:

Distractor Type	English	Hindi
Time-related	15.1%	31.8%
Unrelated	18.7%	35.9%

Some mentions occur coincidentally, so baseline traces without distractors were used for comparison.

Mentioning a time-related distractor is associated with lower odds of a correct answer:

- English: 36% lower odds
- Hindi: 23% lower odds

Key Finding: Hindi reasoning traces incorporate irrelevant information more often, while time-related distractor mentions are associated with reasoning failures in both languages.

Distractor Example

For the short date-addition question that starts “The garden needs new flowers”, a correct Hindi trace ends with

इस तरह, बगीचे में नए फू ल लगाने के लए सही तारीख 2025-03-30 है।

is tarah, bagīce meṃ nae phūl lagāne ke lie sahī tārīkh 2025-03-30 hai.

‘Thus, the correct date for planting new flowers in the garden is 2025-03-30.’

Appendix A.1 relates these mentions to answer correctness. The distractor also leaks into wrong answers. For example, after the distractor “February is such a unique month with its varying days”, a Hindi direct answer to a recurrence question whose correct answer is 2025-04-10 was 2025-02-06. Overall, 12.1% (76 of 629) of wrong Hindi direct answers contain a number, month or weekday taken from the similar distractor.

	Accuracy			Drop	
	None	Sim.	Dis.	Sim.	Dis.
English					
Direct	63.9	63.0	62.7	+0.9	+1.2
CoT	91.6	90.6	91.2	+0.9	+0.4
CoT − direct				+0.0	−0.8
Hindi					
Direct	57.4	58.1	57.7	−0.7	−0.3
CoT	65.2	60.7	62.7	+4.5	+2.5
CoT − direct				**+5.2**	**+2.8**

Error Analysis and Overall Findings (RQ1 + RQ2)

- English CoT errors: 66.5% of incorrect answers are off by one day, suggesting frequent calculation slips.
- Hindi CoT errors: Only 27.1% are off by one day; errors also include incorrect months and reversed date calculations.
- In date recurrence, Hindi counts backward instead of forward in 11.8% of CoT answers, compared to 0.5% in English. Hindi question wording may contribute to this error.
- RQ1 Conclusion: CoT strongly improves English temporal reasoning but provides inconsistent benefits in Hindi, with larger disparities on difficult tasks.
- RQ2 Conclusion: CoT increases Hindi's vulnerability to distractors, especially time-related ones, while English remains relatively robust.

To Do

- Complete the remaining experiments for all difficulty levels.
- Test Hinglish reasoning and check if changing the Hindi date recurrence questions improves the results.
- Test Cross-Lingual Prompting (CLP) and compare it with Hindi CoT.
- Build a parser and label the steps in the model's reasoning.
- Compare English and Hindi reasoning steps and check if their differences lead to incorrect answers.